

Population-Aligned Persona Generation for LLM-based Social Simulation

Zhengyu Hu¹, Jianxun Lian², Zheyuan Xiao¹, Max Xiong³, Yuxuan Lei²,
Tianfu Wang¹, Kaize Ding⁴, Ziang Xiao⁵, Nicholas Jing Yuan⁶, Xing Xie²

¹ HKUST ² Microsoft Research Asia ³ Duke University ⁴ Northwestern University
⁵ Johns Hopkins University ⁶ Microsoft

Abstract

Recent advances in large language models (LLMs) have enabled human-like social simulations at unprecedented scale and fidelity, offering new opportunities for computational social science. A key challenge, however, is the construction of persona sets that authentically represent the diversity and distribution of real-world populations. Most existing LLM-based social simulation studies focus primarily on designing agentic frameworks and simulation environments, often overlooking the complexities of persona generation and the potential biases introduced by unrepresentative persona sets. In this paper, we propose a systematic framework for synthesizing high-quality, population-aligned persona sets for LLM-driven social simulation. Our approach begins by leveraging LLMs to generate narrative personas from long-term social media data, followed by rigorous quality assessment to filter out low-fidelity profiles. We then apply importance sampling to achieve global alignment with reference psychometric distributions, such as the Big Five personality traits. To address the needs of specific simulation contexts, we further introduce a task-specific module that adapts the globally aligned persona set to targeted subpopulations. Extensive experiments demonstrate that our method significantly reduces population-level bias and enables accurate, flexible social simulation for a wide range of research and policy applications.

CCS Concepts

- Computing methodologies → Natural language processing;
- Human-centered computing → Human computer interaction (HCI); Collaborative and social computing.

Keywords

LLM-based Social Simulation, Population-Level Alignment

1 Introduction

Recent advances in Large Language Models (LLMs) have revealed impressive general intelligence across diverse tasks such as mathematics [13, 56], coding [33, 78], and reasoning [21, 77]. Unlike traditional machine learning models, which are limited to narrow domains, LLMs display broad, human-like flexibility [5, 17, 24]. Furthermore, a growing body of research shows that LLMs start to exhibit features of human cognition, including aspects of theory of mind and social intelligence [2, 18, 42, 59, 61, 85]. These capabilities position LLM-based social simulation as a highly promising direction for computational social science. By assigning proper personas and prompting LLMs to role-play as individuals, researchers can simulate human behaviors and attitudes at scale for various applications such as policy analysis [7, 16], behavioral prediction [46, 58],

and fairness assessment [9, 31]. This not only accelerates social science research but also reduces reliance on human subjects for sensitive studies and enables early detection of potential social risks.

A central challenge in simulating societal patterns and dynamics lies in constructing a persona set whose distribution authentically reflects the diversity of traits, behaviors, and psychometric profiles found in real-world populations. However, most large-scale social simulation studies focus primarily on agentic frameworks, often overlooking the complexities and importance of persona set construction. For example, Zhang et al. [84] simply selects 1,000 random Weibo users, [19] uses only three demographic attributes, and [67] relies on LLM-generated or randomly sampled profiles. While [82] adopts a more sophisticated approach, it has two notable limitations: the persona schema is restricted to a fixed set of 15 demographic dimensions, and it assumes that a target group distribution is available prior to persona set construction—an assumption that rarely holds in practice. Overall, if the persona set is not representative, simulation results risk significant bias. This paper aims to provide a rigorous study on constructing faithful persona sets and examines their impact on social simulation outcomes.

As an illustrative example (see Figure 1), we demonstrate how experimental results can be biased when a dataset focuses solely on individual-level persona quality while overlooking alignment at the population level. In this example, we prompt an LLM (Qwen2.5-72B) to role-play as humans completing the IPIP self-report inventory [26] to assess the Big Five personality traits. Figure 1(a) shows the distribution of Big Five traits among real humans, based on data from over one million individuals across 223 countries and regions [47], representing a globally diverse psychometric population. As a naive baseline, Figure 1(b) presents the distribution obtained by running the LLM without any persona assignment and using a high temperature (1.0), which reflects variation due solely to model randomness. We then compare six well-known persona sets from the literature—Tulu-3-Persona [35], Bavard [80], Google Synthetic [27], AlignX [38], Nvidia Nemotron [43], and PersonalHub [4], with their distributions shown in Figure 1(c-h), along with an additional persona set shown in Figure 1(i), which was generated by GPT-4o using the same sample size as the other sets under a high temperature setting (1.0). As expected, while some improvements are observed when persona sets are provided compared to the random baseline, substantial misalignment with real-world distributions persists at the population level. This highlights the urgent need for principled approaches to ensure faithful population-level simulation in LLM-based social modeling.

In this paper, we study the establishment of a synthesized persona foundation for LLM-based social simulation. We propose a

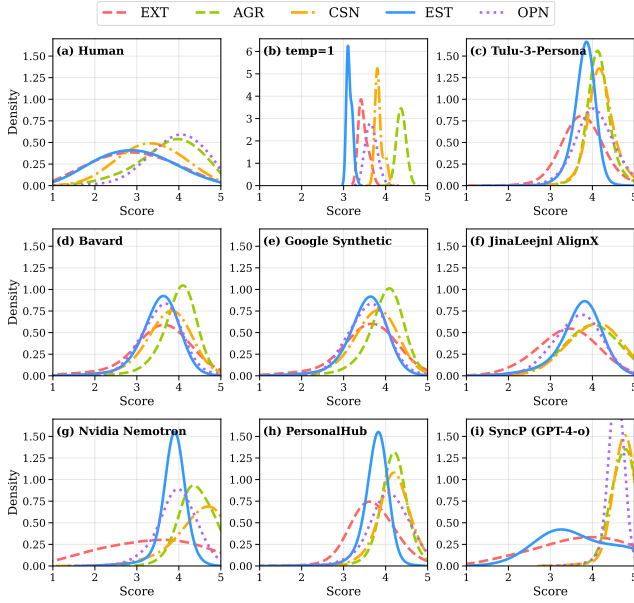


Figure 1: Distributions of Big Five personality traits from real human data, LLM responses without personas, six existing persona sets, and GPT-4o-generated personas.

systematic framework to construct a high-quality persona set that is well-aligned with real-world population distributions. Our framework consists of three key designs:

(1) **High-quality Individual Personas.** First, to ensure that synthesized personas vividly and accurately represent real individuals, we use the Llama-3.3-70B model to extract narrative personas from users’ long-term social media histories, drawing data from platform: blogs [54]. Unlike previous works [19, 82] that use a fixed set of demographic attributes as the persona schema, we opt for narrative personas to retain flexibility and avoid inaccuracies stemming from missing demographic information. We then leverage another LLM, Qwen2.5-72B, as an evaluator to rate the quality of synthesized personas, retaining only those that exceed a preset quality threshold. Experimental results demonstrate that this generated persona set already significantly outperforms existing baselines in capturing population-level distributions across various psychometric measures.

(2) **Global Distribution Alignment.** Despite these improvements, the initially generated persona set still contains biases introduced by the unique user characteristics of online social platform. In the second design, we prompt each raw persona to complete a psychometric inventory via the LLM, producing an *LLM-based* response distribution. We then compare this distribution with the human reference and employ a two-stage resampling techniques, importance sampling followed by optimal transport, to make resampled distribution matches the human one. The personas linked to these resampled points constitute our population-aligned persona set. Given the generalizability of the Big Five personality traits, we adopt the widely used IPIP Big Five psychometric test [20, 26, 47] as the reference for aligning personality trait distributions across the global population. To assess the effectiveness of this alignment,

we further evaluate the selected persona set on several other psychometric tests, such as CFCS [60], FBPS [48], and Duckworth [14], as out-of-domain (OOD) benchmarks. Our experimental results confirm that the global alignment generalizes well to these OOD tests.

(3) **Group-specific Population Adjustment.** Recognizing that most real-world social simulation applications do not require modeling the global population but instead focus on specific groups, such as college students or residents of a particular country, our third design provides a module for generating group-specific persona sets. **Given a task-specific query, this module retrieves relevant personas from the globally aligned set using embedding similarity to form a seed set.** We then employ an LLM to make minor revisions to each persona in the seed set, ensuring their suitability for the current task. Experimental results on YRBSS [3] and regional WVS [25] datasets demonstrate that this approach effectively produces group-specific persona populations.

In summary, our contributions include:

- **We emphasize the importance of population-level persona alignment for LLM-based social simulation, and propose a systematic framework to generate persona sets that accurately reflect real-world population distributions.**
- **We design a two-stage alignment method combining Importance Sampling and Optimal Transport for global distribution matching.**
- **We develop a persona targeting module that supports group-specific persona alignment through embedding retrieval and LLM revision, thereby facilitating its practical application in diverse social science scenarios.**
- **Through comprehensive evaluation across six different psychometric tests, we demonstrate the superiority of our framework over existing persona sets in terms of distributional alignment, behavioral consistency, and group-specific alignment scenarios.**

2 Related Work

LLM-based Simulation for Social Science. LLMs are increasingly employed to simulate human behavior, presenting new possibilities for social science research [10, 19, 49, 67, 68, 71, 74, 75, 82, 84]. These simulations enable scalable and cost-effective analyses of attitudes, preferences, and decision-making processes [12, 36, 70, 74]. Prior work has primarily emphasized individual-level fidelity, focusing on ensuring that simulated agents produce realistic and consistent outputs. For instance, Shao et al. [57] simulate users in dialogue tasks to enable personalized conversation generation, while Kong et al. [34] study persona style transfer to increase dialogue diversity. Other studies have explored modeling individual reasoning traits [6], proposing persona-aware evaluation metrics [41], and introducing benchmarks such as CharacterEval for role-playing agent evaluation [63]. Group-level phenomena have also been examined, including polarization dynamics [50], partisan group behaviors [11], and role-playing ordinary individuals [44]. Recently, Wu et al. [73] point out that LLM-based simulations tend to default to an “average persona” that underrepresents behavioral heterogeneity, thereby diminishing the realism of emergent group dynamics. They argue that focusing on collective patterns can mitigate these limitations

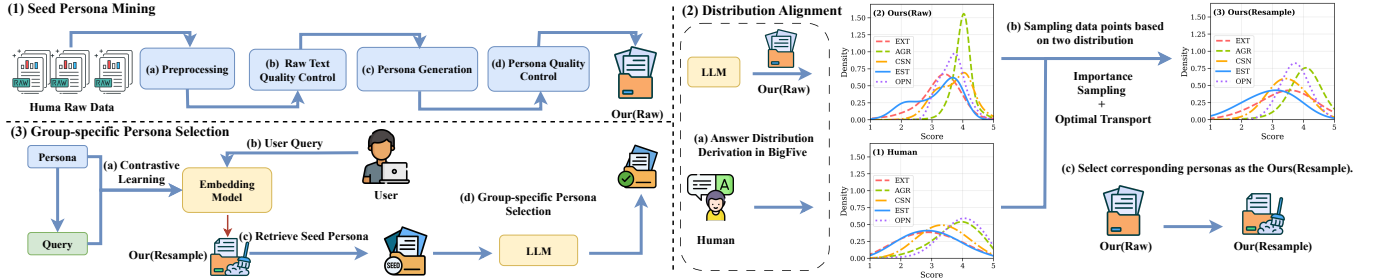


Figure 2: Overview of the Population-Aligned Persona Framework.

by anchoring simulations in population-level distributions. However, existing approaches often lack mechanisms to ensure that simulated persona populations collectively reflect real-world trait distributions, which can result in biased simulation outcomes.

Persona Modeling and Evaluation. Personas are widely utilized in NLP for modeling agent traits and behaviors in dialogue [22, 72], user modeling [32, 39], and social simulation [8, 23, 29]. Persona generation methods range from manual curation to automatic approaches using LLMs. Early work introduced persona-based neural conversation models to encode speaker traits for consistent dialogue [37], while later studies enhanced engagement by conditioning responses on both self and partner profiles [81]. More recent methods employ LLMs to simulate historical or fictional characters [57] and to guide profile generation for personalized outputs [79]. Evaluation has largely focused on individual-level attributes such as coherence, consistency, and persona-grounded dialogue. Recent benchmarks and frameworks, including RPG-Bench [76], PersonaGym [53], and analyses of lexical diversity in persona descriptions [55], assess the role-playing and fidelity of LLM-generated personas. While these efforts advance persona-based simulation and individual agent fidelity, comprehensive frameworks for constructing population-aligned persona sets remain scarce, leading to meaningful biases in simulation outcomes. Our work addresses this gap by introducing a principled pipeline that generates narrative personas, aligns persona sets with psychometric reference distributions (e.g. Big Five), and adapts persona populations to task-specific subgroups.

3 Population-Aligned Persona Framework

Our framework consists of three key components: (1) *Seed Persona Mining*, which extracts and filters high-quality narrative personas from large-scale human-authored corpora; (2) *Global Distribution Alignment*, which uses a two-stage sampling strategy, Importance Sampling and Optimal Transport, to align persona distributions with human psychometric data; and (3) *Group-specific Persona Construction*, where we support targeted persona generation for specific tasks. An overview is shown in Figure 2.

3.1 Seed Persona Mining

We construct a high-quality seed persona pool from a large-scale human-authored corpora: the *Blog Authorship Corpus* [54] (681K posts). This sources provide diverse narratives across demographic and behavioral dimensions. The mining pipeline includes four stages:

Preprocessing. We clean each corpus to ensure data authenticity and narrative richness. For the blog corpus, we discard entries under 30 tokens or lacking first-person pronouns. All remaining texts are rewritten using Llama-3-70B to remove HTML tags, URLs, non-English content, and formatting noise. This results in approximately 500K blog.

Raw Text Quality Control. Each rewritten post is annotated using Llama-3-70B with two labels: a three-level quality score (high, medium, low), and a binary harmfulness label. High-quality content is expected to reflect personal experience, emotions, or social behavior, while excluding boilerplate or ad-like content. Harmful content (e.g., violence, privacy violations) is filtered out. Only posts rated both high-quality and harmless are retained, resulting in final counts of 368K blog.

Persona Generation. We aggregate posts by author and discard users with fewer than six posts. Each user’s posts are concatenated into a single text block s , which is passed to Llama-3-70B to generate a grounded persona summary $p_i = \mathcal{G}(s)$, where the model is prompted to cite evidence from s for factual grounding.

Persona Quality Control. To ensure quality, we use a critic LLM to score each persona p_i on five dimensions: hallucination (faithfulness to source text), coverage (salience of user traits), conciseness (well-formed and within 100 words), relevance (exclusion of generic or promotional content), and overall quality. Each score ranges from 1 to 10. We retain personas with an overall score >8 and all other dimensions >7 . This step filters out approximately 30% of personas from each source.

The resulting seed pool contains over 160,000 high-fidelity narrative personas covering diverse psychometric and demographic attributes. Full data statistics, prompt templates, and filtering thresholds are detailed in Appendix F.

3.2 Global Distribution Alignment via Two-stage Sampling

Let $\mathcal{P} = \{p_i\}_{i=1}^N$ denote a set of high-quality textual personas (see Section 3.1), and let $\mathcal{Q} = \{q_k\}_{k=1}^d$ be a fixed set of psychometric prompts (e.g., Big Five questionnaire items). Let LLM be a language model that maps a concatenated persona-question pair to a scalar response. For each persona p_i and question q_k , we define the model response as:

$$x_{ik} := \text{LLM}(p_i \parallel q_k), \quad (1)$$

where $\parallel$ denotes textual concatenation. We collect the responses for all d questions to form a response vector $x_i = (x_{i1}, \dots, x_{id}) \in \mathbb{R}^d$

for each persona p_i . The full set $\{x_i\}_{i=1}^N$ defines an empirical distribution over model-induced responses, which we denote as r_{persona} . Similarly, we let $Y = \{y_j\}_{j=1}^M \subset \mathbb{R}^d$ represent the human response vectors obtained from survey data, and denote the corresponding empirical distribution as r_{human} . In this work, we use large-scale human response data from the widely used IPIP Big Five inventory [20, 26, 47], spanning one million individuals across 223 countries and regions, as a representative reference for real-world personality distributions.

Given a persona set $\mathcal{P}$ with induced response distribution r_{persona} and a human response distribution r_{human} , our objective is to select a subset $\mathcal{P}' \subseteq \mathcal{P}$, such that its empirical distribution recovers the human distribution. To achieve this, we adopt a two-stage sampling framework combining Importance Sampling (IS) and Optimal Transport (OT). The IS stage acts as a coarse filter that concentrates sampling on regions where the persona distribution overlaps significantly with the human distribution, thereby reducing both computational cost and the risk of OT degenerating due to large distributional gaps. The subsequent OT stage then performs fine-grained alignment within this candidate pool, ensuring that the final sampled personas closely match the detailed structure of human responses.

Stage 1: Importance Sampling (IS). In the first stage, our goal is to efficiently select a subset of personas whose response distributions significantly overlap with the human distribution. To achieve this, we estimate the probability densities of both the human response distribution r_{human} and the original persona distribution r_{persona} via Kernel Density Estimation (KDE). Specifically, employing a Gaussian kernel with bandwidth $h > 0$, we obtain the density estimates:

$$\begin{aligned}\hat{r}_{\text{human}}(x) &= \frac{1}{M} \sum_{j=1}^M \mathcal{N}(x - y_j; h^2 I), \\ \hat{r}_{\text{persona}}(x) &= \frac{1}{N} \sum_{i=1}^N \mathcal{N}(x - x_i; h^2 I),\end{aligned}\quad (2)$$

where $I \in \mathbb{R}^{d \times d}$ denotes the identity matrix, and $\mathcal{N}(z; h^2 I)$ represents a multivariate Gaussian density centered at zero with covariance $h^2 I$. Using these KDE-based estimates, we calculate an importance weight for each persona response vector x_i as the ratio of the estimated densities: $w_i^{\text{IS}} = \frac{\hat{r}_{\text{human}}(x_i)}{\hat{r}_{\text{persona}}(x_i)}$. Intuitively, the weight w_i^{IS} quantifies the relative likelihood of observing the response vector x_i under the human response distribution compared to the original persona distribution. Therefore, personas with higher importance weights indicate a stronger alignment with human responses. To convert these importance weights into valid sampling probabilities, we normalize them as:

$$\pi_i = \frac{w_i^{\text{IS}}}{\sum_{k=1}^N w_k^{\text{IS}}}, \quad \text{with } \sum_{i=1}^N \pi_i = 1. \quad (3)$$

Finally, we draw $N^\dagger$ candidate personas from the original set $\mathcal{P}$ according to the probabilities $\{\pi_i\}$:

$$\mathcal{P}^\dagger \sim \text{Multinomial}\left(N^\dagger, \{\pi_i\}_{i=1}^N\right), \quad (4)$$

forming the candidate set $X^\dagger = \{x_i \mid p_i \in \mathcal{P}^\dagger\}$. This step effectively reduces the sample space to personas closely resembling human respondents, providing a high-quality initial candidate pool for further refinement in the next stage.

Stage 2: Optimal Transport (OT). While Importance Sampling identifies a promising subset of personas, it does not guarantee fine-grained alignment with the detailed structure of the human response distribution. Therefore, we introduce a second stage leveraging entropic Optimal Transport to precisely match the selected personas to the human data at a granular level.

Recall that each persona response vector $x_i^\dagger \in \mathbb{R}^d$ and human response vector $y_j \in \mathbb{R}^d$ comprise responses to a set of d psychometric prompts $Q = \{q_k\}_{k=1}^d$. To explicitly account for the importance of individual questionnaire items during alignment, we introduce a set of weights $\{\omega_k\}_{k=1}^d$, with $\omega_k > 0$ denoting the importance of prompt q_k . Thus, we construct the cost matrix $C \in \mathbb{R}^{N^\dagger \times M}$ as a weighted sum of squared differences:

$$C_{ij} = \sum_{k=1}^d \omega_k \left(x_{ik}^\dagger - y_{jk}\right)^2. \quad (5)$$

Our goal is then to identify an optimal probabilistic matching (transport plan) $\Gamma \in \mathbb{R}^{N^\dagger \times M}$ between the two sets of vectors, such that the total weighted transport cost is minimized while maintaining uniform marginal distributions. We formulate the following entropic-regularized OT optimization problem:

$$\Gamma^\star = \arg \min_{\Gamma \in \Pi(\mathbf{a}^\dagger, \mathbf{b})} \langle C, \Gamma \rangle + \varepsilon \sum_{i,j} \Gamma_{ij} (\log \Gamma_{ij} - 1), \quad (6)$$

where $\varepsilon > 0$ is a regularization parameter that controls the entropy of the resulting transport plan. Each entry $\Gamma_{ij} \in \mathbb{R}_{\geq 0}$ denotes the fraction of probability mass assigned to matching persona $x_i^\dagger$ with human response y_j , with higher values indicating closer alignment.

We define the marginals $\mathbf{a}^\dagger \in \mathbb{R}^{N^\dagger}$ and $\mathbf{b} \in \mathbb{R}^M$ as uniform distributions over the candidate personas and the real human samples, respectively:

$$\mathbf{a}^\dagger = \frac{1}{N^\dagger} \mathbf{1}_{N^\dagger}, \quad \mathbf{b} = \frac{1}{M} \mathbf{1}_M, \quad (7)$$

where $N^\dagger$ is the number of candidate personas after importance sampling, and M is the number of empirical human response vectors from survey data. Here, $\mathbf{1}_M \in \mathbb{R}^M$ and $\mathbf{1}_{N^\dagger} \in \mathbb{R}^{N^\dagger}$ are vectors of all ones. Given these marginals, the feasible set $\Pi(\mathbf{a}^\dagger, \mathbf{b})$ enforces marginal constraints on the transport plan:

$$\Pi(\mathbf{a}^\dagger, \mathbf{b}) = \left\{ \Gamma \in \mathbb{R}_{\geq 0}^{N^\dagger \times M} \mid \Gamma \mathbf{1}_M = \mathbf{a}^\dagger, \Gamma^\top \mathbf{1}_{N^\dagger} = \mathbf{b} \right\}. \quad (8)$$

The constraint $\Gamma \mathbf{1}_M = \mathbf{a}^\dagger$ ensures that each persona distributes its probability mass across all human samples, while $\Gamma^\top \mathbf{1}_{N^\dagger} = \mathbf{b}$ ensures that each human sample receives the correct total mass.

To solve the OT problem efficiently, we employ the Sinkhorn-Knopp iterative algorithm. Specifically, we first define the Gibbs kernel $K \in \mathbb{R}^{N^\dagger \times M}$ by exponentiating the negative normalized cost matrix:

$$K_{ij} = \exp\left(-\frac{C_{ij}}{\varepsilon}\right), \quad (9)$$

where each K_{ij} represents the unnormalized probability mass assigned to transporting from persona i to human j , inversely weighted

by the cost. Let $u \in \mathbb{R}^{N^\dagger}$ and $v \in \mathbb{R}^M$ be scaling vectors that normalize rows and columns of the transport plan, respectively. The iterative updates are:

$$u \leftarrow \frac{\mathbf{a}^\dagger}{Kv}, \quad v \leftarrow \frac{\mathbf{b}}{K^\top u}, \quad (10)$$

where Kv denotes matrix-vector multiplication, yielding a vector in $\mathbb{R}^{N^\dagger}$ whose i -th element is $\sum_j K_{ij}v_j$, and similarly $K^\top u \in \mathbb{R}^M$ denotes the column-wise normalization step. These updates ensure that the resulting plan Γ satisfies the marginal constraints. At convergence, we obtain the optimal transport plan Γ^* as:

$$\Gamma^* = \text{diag}(u) \cdot K \cdot \text{diag}(v), \quad (11)$$

where $\text{diag}(u)$ and $\text{diag}(v)$ denote diagonal matrices with u and v on their diagonals, respectively. Finally, we compute a cost-aware OT importance weight for each candidate persona $x_i^\dagger$ by taking the expected transport cost under Γ^* and mapping it through a temperature-controlled exponential:

$$a_i^\dagger = \sum_{j=1}^M \Gamma_{ij}^*, \quad w_i^{\text{OT}} = \exp\left(-\frac{1}{\tau a_i^\dagger} \sum_{j=1}^M \Gamma_{ij}^* C_{ij}\right), \quad (12)$$

where $\tau > 0$ is a temperature hyperparameter. After normalization, we sample the final aligned subset $\mathcal{P}' \subseteq \mathcal{P}^\dagger$ of size N' :

$$\pi_i^{\text{OT}} = \frac{w_i^{\text{OT}}}{\sum_{k=1}^{N^\dagger} w_k^{\text{OT}}}, \quad \mathcal{P}' \sim \text{Multinomial}\left(N', \{\pi_i^{\text{OT}}\}_{i=1}^{N^\dagger}\right). \quad (13)$$

This stage ensures fine-grained alignment with human distributions by weighting prompt importance and applying optimal transport for precise persona selection.

Discussion. Beyond empirical results, we establish finite-sample guarantees showing that our two-stage procedure provably recovers the target human distribution. In particular, we bound (i) the Wasserstein-2 error introduced by the KDE-based importance sampling stage and (ii) the additional $O(\varepsilon)$ bias from the entropic OT refinement. These results imply that the resampled persona distribution converges to the human distribution at a rate that scales with the sample size and kernel bandwidth, providing formal support for the population-level fidelity of our method. Complete theorem statements and proofs are deferred to Appendix G.

3.3 Group-specific Persona Set Construction

To support adaptive construction of group-specific persona subsets, we train a dedicated embedding model that embeds natural language queries and personas into a shared semantic space. For training data construction, we prompt a LLM¹ to generate descriptive queries from each persona $p_i \in \mathcal{P}'$:

$$q_i = \text{LLM}(p_i), \quad (14)$$

where each q_i is a natural language description semantically aligned with p_i . These (q_i, p_i) pairs serve as positive samples. To construct negative samples, we first compute similarity between q_i and all other personas, and select the top- N most similar candidates. We also sample N random personas. Since both the top- N most similar candidates and the N randomly sampled personas may contain false negatives (i.e., personas that actually match the query), we further

use an LLM to filter out any candidates that are semantically aligned with the query. The embedding model is trained using following loss:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(e_q, e_p^+))}{\exp(\text{sim}(e_q, e_p^+)) + \sum_{p^-} \exp(\text{sim}(e_q, e_{p^-}))}, \quad (15)$$

where e_q and e_p are the embeddings of the query and persona respectively, and $\text{sim}(\cdot, \cdot)$ denotes cosine similarity.

At inference time, given a user query q (e.g., “first-year college students in the United States”), we compute its embedding $e_q = \text{Embed}(q)$ and measure similarity against all persona embeddings $\{e_i\}$ in $\mathcal{P}'$. We then select the top- K most similar personas based on cosine similarity to form a seed subset $\mathcal{P}_{\text{seed}}(q)$. This subset provides semantically relevant exemplars for further persona generation. To generate high-fidelity group-specific personas, we prompt a backbone LLM using the original query and the retrieved seed subset. Specifically, we format the input as a concatenation of q and the selected personas in $\mathcal{P}_{\text{seed}}(q)$, and generate new samples as follows:

$$\mathcal{P}_{\text{group}}(q) = \text{LLM}(q \parallel \mathcal{P}_{\text{seed}}(q)). \quad (16)$$

The resulting set $\mathcal{P}_{\text{group}}(q)$ aligns with user intent, making it applicable to downstream social simulations constrained by specific demographic or behavioral parameters.

4 Experiments Setup

Implementation Details. In Section 3.3, we set $N = 10$. For both Section 4.1 and Section 4.2, we adopt a unified setup. In the *Without Persona* setting, each base model answers the questionnaire directly under three temperature settings (0.0, 0.7, 1.0), generating 5,000 times per temperature. In the *With Persona* setting, all persona sets are standardized to 5,000 personas. For SyncP-type methods (e.g., SyncP (GPT-4-o)), we generate 5,000 personas using GPT-4-o at temperature 1.0. For public sets (Tulu-3-Persona [35], Bavard [80], Google Synthetic [27], AlignX [38], Nvidia Nemotron [43], and PersonalHub [4]), we randomly sample 5,000 personas from each. Our methods, Ours (RandomSelect) and Ours (Resample), also use 5,000 personas drawn from our high-quality seed pool, with Resample applying the full alignment pipeline described in Section 3.2. The experimental setup in Section 4.3 follows the same in-domain configuration as described in Section 4.1. The embedding model used in Section 4.4 is Qwen3-Embedding-0.6B [83].

Baselines. To evaluate the effectiveness of our framework, we compare against two categories of baselines: *Without Persona* and *With Persona*. The *Without Persona* setting uses a LLM to answer tasks directly without any persona conditioning, serving as a reference for understanding the impact of persona guidance. We include **DAns** under three temperature settings, **temp=0, 0.7, and 1** to capture the effect of sampling variability. The *With Persona* category incorporates persona-augmented prompting. We first consider **SyncP**, which generates personas using the corresponding model; for example, **SyncP (GPT-4-o)** denotes personas generated by GPT-4-o at temperature 1. In addition to SyncP, we incorporate several widely adopted, publicly available persona sets as baselines: **Tulu-3-Persona** [35], **Bavard** [80], **Google Synthetic** [27], **AlignX** [38], **Nvidia Nemotron** [43], and **PersonalHub** [4]. To assess the impact of our alignment pipeline, we introduce two variants of our

¹All LLMs used in this section are Qwen2.5-72B.

Table 1: In-domain alignment results on the IPIP Big Five dataset across three base models across four metrics Lower is better for all metrics. The minimum value in each column is highlighted in bold.

Model Base		Qwen2.5-72B				Llama-3-70B				Phi-4				Avg.
		AMW	FD	SW	MMD	AMW	FD	SW	MMD	AMW	FD	SW	MMD	
Without Persona	DAns (temp=0)	0.4111	1.4256	0.4112	0.8631	0.6469	3.4211	0.6962	2.1904	0.4560	1.7358	0.4602	1.1207	1.1532
	DAns (temp=0.7)	0.3860	1.2309	0.3742	0.6969	0.6271	3.0539	0.6611	1.8805	0.3909	1.3098	0.3960	0.8257	0.9861
	DAns (temp=1)	0.3687	1.0947	0.3560	0.6250	0.5946	2.9176	0.6422	1.7867	0.3731	1.1475	0.3744	0.7294	0.9175
With Persona	SyncP (Qwen2.5-72B)	0.2639	0.5854	0.2792	0.3193	0.3124	0.8279	0.3102	0.4749	0.3284	0.8232	0.3541	0.4282	0.4423
	SyncP (Llama-3-70B)	0.2759	0.5832	0.2679	0.3392	0.3996	0.8307	0.3472	0.4421	0.3336	0.8268	0.3383	0.4886	0.4561
	SyncP (Phi-4)	0.2850	0.7975	0.3363	0.4316	0.3884	1.1620	0.4011	0.7135	0.3357	0.9425	0.3609	0.5767	0.5609
	SyncP (GPT-4-o)	0.2467	0.5824	0.2679	0.2701	0.3523	0.7263	0.3387	0.4476	0.3249	0.7500	0.3175	0.4147	0.4199
	Tulu-3-Persona	0.2821	0.8400	0.3250	0.4631	0.3276	0.9418	0.3512	0.5941	0.2814	0.6412	0.2788	0.3593	0.4738
	Bavard	0.3069	0.7838	0.3174	0.4669	0.3173	0.6659	0.3476	0.3758	0.1877	0.3759	0.2028	0.2414	0.3825
	Google Synthetic	0.3135	0.8081	0.3246	0.4880	0.2983	0.6715	0.3286	0.3684	0.1878	0.3729	0.2027	0.2394	0.3837
	AlignX	0.3416	0.9393	0.3567	0.5606	0.3319	0.6772	0.3274	0.3481	0.1641	0.3215	0.2163	0.2306	0.4013
	Nvidia Nemotron	0.2645	0.8316	0.3199	0.4414	0.3615	0.8005	0.3177	0.4497	0.2233	0.5119	0.2412	0.2421	0.4171
	PersonalHub	0.2982	0.9167	0.3436	0.5303	0.3365	0.8011	0.3268	0.5001	0.2469	0.5620	0.2609	0.3112	0.4529
Ours	Raw	0.2178	0.4430	0.2246	0.2315	0.2528	0.5597	0.2688	0.3302	0.1181	0.2373	0.1429	0.0689	0.2580
	RandomSelect	0.2178	0.4416	0.2249	0.2311	0.2523	0.5595	0.2699	0.3291	0.1190	0.2366	0.1430	0.0680	0.2577
	Resample	0.1527	0.2628	0.1272	0.0840	0.2013	0.4036	0.1955	0.1937	0.0889	0.1911	0.1214	0.0353	0.1715

method. **Ours (Raw)** uses the full persona set from Section 3.1 without any sampling. Based on this pool, **Ours (RandomSelect)** randomly samples a subset matching the size of other baselines, serving as a size-controlled reference. **Ours (Resample)** applies our full pipeline in Section 3.2 to select a distributionally aligned subset of the same size. This setup allows us to isolate the effect of alignment from data source and sample size. In Section 4.4, we further evaluate two additional group-specific variants. **Ours (EM w/o train)** retrieves from **Ours (Resample)** using an untrained embedding model, while **Ours (EM w train)** uses a contrastively trained embedding model to retrieve semantically relevant personas. The embedding model is trained with positive and filtered hard-negative pairs (see Section 3.3).

Evaluation Datasets. This paper using six different psychometric tests: the IPIP Big Five psychometric test [26], CFCS [60], FBPS [48], Duckworth [14], WVS [25], and YRBSS [3]. Detailed descriptions are provided in Appendix B.2.

Evaluation Metrics. To comprehensively evaluate the quality of LLM-based social simulation, we adopt two complementary classes of metrics that assess both distribution-level and response-level fidelity. For population-level alignment, we compare the overall distributions between synthetic and human responses using four standard metrics: Averaged Monotonic Wasserstein (AMW) [65, 66], Fréchet Distance (FD) [28, 64], Sliced Wasserstein Distance (SW) [40, 45], and Maximum Mean Discrepancy (MMD) [30, 52], which capture discrepancies across both marginal and joint distributions. For individual-level consistency, we compute Pearson correlation coefficients across all trait pairs and report the mean absolute error (MAE) between synthetic and human correlations, following [1, 51]. Formal definitions of all metrics are provided in Appendix D.

4.1 Population Level

In-domain results. Table 1 reports in-domain alignment results across three open-source LLMs: Qwen2.5-72B, Llama-3-70B, and Phi-4. The task evaluates how closely model outputs match human Big Five trait distributions from the IPIP inventory, which also serves as the reference distribution in our alignment pipeline. In the *Without Persona* setting, increasing the sampling temperature from 0 to 1 improves alignment across all models—for instance, FD on Qwen2.5-72B drops from 1.43 to 1.09, indicating that decoding randomness introduces more human-like variation. However, even at optimal temperature, these settings perform far worse than persona-conditioned baselines, underscoring the limitations of unconstrained generation. All *With Persona* baselines yield substantial improvements. Public persona sets (e.g., Tulu-3-Persona, AlignX, PersonalHub) moderately reduce alignment error, while SyncP variants (e.g., GPT-4-o-generated personas) perform comparably but do not consistently surpass hand-crafted sets. These gains confirm the value of personas, but also reveal persistent mismatches with population-level ground truth, highlighting the limits of naive persona generation. Our proposed framework delivers the strongest and most consistent alignment across all models. The RESAMPLE variant—built via our full pipeline of importance sampling and optimal transport—achieves the lowest average error in every case, reducing alignment error by 49.8% on Qwen2.5-72B (vs. SyncP), 37.9% on Llama-3-70B (vs. Google Synthetic), and 45.1% on Phi-4 (vs. AlignX). These consistent gains validate the effectiveness of our alignment strategy across architectures. Even our unaligned variants (RAW and RANDOMSELECT) outperform most public persona sets, demonstrating the strength of our seed pool. Their near-identical performance also confirms that the gains from RESAMPLE stem from distributional alignment rather than sample size effects.

Out-of-domain results. Table 2 presents the population-level alignment results on three psychological instruments, CFCS [60], FBPS [48], and Duwo i.e., Duckworth [14], that were not used during persona

Table 2: Out-of-domain alignment results on CFCS, FBPS, and Duckworth datasets across four metrics Lower is better for all metrics. The minimum value in each column is highlighted in bold.

	CFCS				FBPS				Duckworth				Avg.
	AMW	FD	SW	MMD	AMW	FD	SW	MMD	AMW	FD	SW	MMD	
AlignX	0.2189	0.0697	0.2189	0.1091	0.3527	1.0808	0.3122	0.5710	0.4083	0.4333	1.2429	0.7111	0.4774
Tulu-3-Persona	0.2755	0.0826	0.2755	0.1450	0.2418	0.4849	0.2716	0.2564	0.3838	0.4009	0.7533	0.5815	0.3461
Bavard	0.2415	0.0870	0.2415	0.1171	0.2773	0.6788	0.2545	0.3717	0.3048	0.3052	0.7038	0.4301	0.3344
Google Synthetic	0.2563	0.0972	0.2563	0.1320	0.2857	0.7112	0.2648	0.3985	0.3141	0.3122	0.7295	0.4524	0.3509
Nvidia Nemotron	0.2514	0.1029	0.2514	0.1178	0.2595	0.9599	0.2817	0.5387	0.2970	0.4054	1.1136	0.6837	0.4386
SyncP (GPT-4-o)	0.3258	0.1591	0.3258	0.2043	0.3066	0.9925	0.3054	0.5456	0.3287	0.3573	0.8842	0.5241	0.4383
PersonalHub	0.2099	0.0975	0.2099	0.0885	0.2453	0.6068	0.2286	0.3075	0.3236	0.3664	0.9190	0.5443	0.3456
Ours (Resample)	0.1640	0.0303	0.1648	0.0565	0.2149	0.4004	0.1814	0.2005	0.2340	0.2235	0.3860	0.2453	0.2085

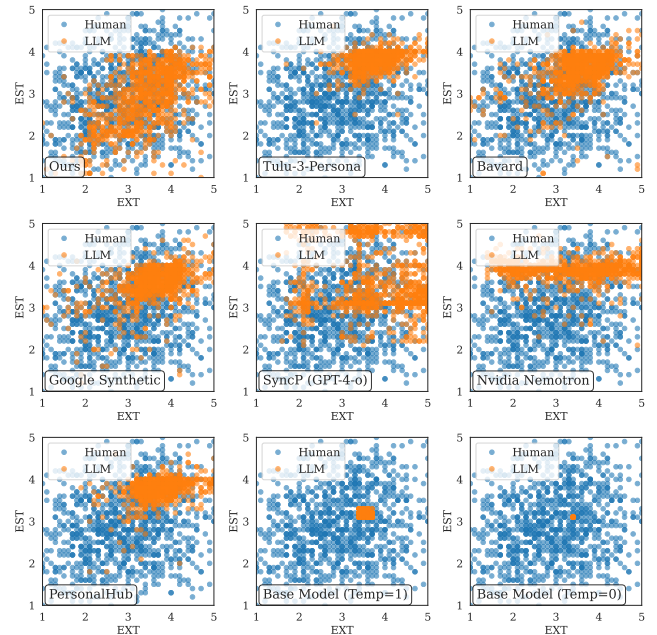
construction. Despite measuring distinct constructs, these datasets share an important property: they are all built on globally distributed human responses, covering 143, 168, and 110 countries respectively. This aligns well with the IPIP Big Five inventory used in our alignment process, which spans one million individuals across 223 countries and serves as a proxy for global human trait distribution. Across all three tasks and four evaluation metrics (AMW, FD, SW, MED), RESAMPLE achieves the lowest average divergence (0.2279), outperforming the strongest baseline (Bavard, 0.3344) by 32%. On FBPS our method reduces the Fréchet Distance to 0.4220, substantially outperforming Tulu-3-Persona (0.4849), Google Synthetic (0.7112), and SyncP (GPT-4-o) (0.9925). The additional gains from the alignment process validate the necessity of principled population-level sampling in constructing accurate and transferable persona sets.

Table 3: MAE_{corr} between human and synthetic inter-trait correlations. Lower values indicate better individual-level consistency. The minimum value in each column is highlighted in bold.

Method	BIG5	CFCS	FBPS	Duwo	Avg.
AlignX	0.5761	0.7944	0.5078	0.5688	0.6118
Tulu-3-Persona	0.4971	1.0521	0.4017	0.4970	0.6120
Bavard	0.4528	0.4066	0.4268	0.4559	0.4355
Google Synthetic	0.4570	0.4382	0.4168	0.4571	0.4423
Nvidia Nemotron	0.8000	1.1935	0.6894	0.8075	0.8726
SyncP (GPT-4-o)	1.0412	1.2794	0.9289	1.0443	1.0735
PersonalHub	0.6547	0.9334	0.5045	0.6564	0.6873
Ours (Resample)	0.4108	0.2270	0.3607	0.4255	0.3560

4.2 Individual-Level Behavioral Consistency

Beyond matching trait distributions, realistic simulation must preserve the internal structure of personality traits [69]. We evaluate this using mean absolute error (MAE_{corr}) between pairwise Pearson correlations in human vs. synthetic data (see Section 4). Table 3 reports results across four instruments: IPIP Big Five [26], CFCS [60], FBPS [48], and Duckworth [14]. Our proposed RESAMPLE method consistently achieves the strongest performance, producing personas whose inter-trait correlations most closely mirror those of

**Figure 3: Scatter plots comparing the distribution of EXT and EST scores between real human responses (blue) and LLM-generated outputs (orange) across nine settings.**

real human populations. It attains the lowest error on *every* instrument, with an overall MAE_{corr} of 0.3651, outperforming the strongest prior baseline (Bavard, 0.4355) by 16.2%, and surpassing commonly used sets like AlignX (0.6118) and Tulu-3-Persona (0.6120). These results demonstrate that our alignment pipeline sharpens individual-level consistency, not just population-level fit. Similarly, we observe that the full two-stage alignment procedure provides clear improvements even over our own high-quality seed pool.

4.3 Trait-Space Visualization

To complement our quantitative analysis, we provide a qualitative visualization of personality trait coverage. Following the same in-domain setup as in Section 4.1, we plot the joint distribution

Table 4: Performance of group-specific persona generation across datasets and regions. Lower is better for all metrics. The minimum value in each column is highlighted in bold.

	YRBSS				WVS (East Asia)				WVS (North America)				WVS (Western Europe)				Avg.
	AMW	FD	SW	MMD	AMW	FD	SW	MMD	AMW	FD	SW	MMD	AMW	FD	SW	MMD	
AlignX	0.4676	9.7132	0.5679	2.0409	0.8274	21.0512	0.8712	2.4598	0.8785	20.4461	0.8914	2.4495	0.8595	23.4058	0.8576	2.0302	5.6136
Tulu-3-Persona	0.4921	9.0356	0.7252	1.9259	0.9282	23.2265	0.9343	2.6507	0.8498	20.8217	0.8976	2.6438	0.9485	26.8801	0.9615	2.3601	6.0176
Bavard	0.4569	9.0103	0.5558	1.8776	0.8153	22.7143	0.8190	2.1368	0.8489	20.3607	0.8730	2.4311	0.8991	24.8798	0.8593	1.9787	5.7198
Google Synthetic	0.4716	9.0975	0.5500	1.8997	0.8197	21.5079	0.8443	2.2662	0.8412	22.6565	0.8757	2.6811	0.8818	24.8154	0.8667	1.9872	5.8164
Nvidia Nemotron	0.5052	10.9920	0.6999	2.2383	0.8845	24.7850	0.8990	2.6935	0.8687	20.2640	0.8733	2.4560	0.8793	26.9882	0.8832	2.1464	6.1910
SyncP (GPT-4-o)	0.5473	11.5000	0.7013	2.4586	0.9706	31.6886	0.9890	3.4136	0.8183	22.5895	0.8881	2.4783	1.0129	31.3838	1.0204	3.0113	7.2170
PersonalHub	0.4834	9.9214	0.6021	2.1532	0.8144	22.8832	0.8262	2.2022	0.8204	21.1220	0.8551	2.7758	0.8678	24.9372	0.8577	1.9928	5.8822
Ours (EM w/o train)	0.2940	2.5384	0.3338	0.7464	0.8011	20.6770	0.8032	2.1235	0.7957	20.1950	0.8386	2.2835	0.8286	22.0073	0.8362	1.8213	4.8702
Ours (EM w train)	0.2819	2.4790	0.3173	0.7324	0.7713	19.4662	0.7758	1.9500	0.7694	17.5770	0.7764	1.7992	0.7858	21.5000	0.8108	1.7339	4.5329

over two representative Big Five traits: **Extraversion (EXT)** and **Emotional Stability (EST)**. Figure 3 shows scatter plots of 1,000 samples per setting across nine representative conditions, where the x -axis denotes EXT and the y -axis denotes EST. Blue points represent real human responses; orange points denote LLM outputs. Each subplot is labeled with the corresponding method. In the **Temp = 0** setting (bottom right), all responses collapse to a single point due to deterministic decoding. Increasing temperature to 1 adds some dispersion, but the outputs remain narrowly clustered, failing to reflect real-world variability. This shows that sampling alone is insufficient for simulating population-level diversity. Persona-based approaches (e.g., Tulu-3-Persona, Bavard, Google Synthetic) offer modest improvements, with outputs spreading further into trait space and partially overlapping human data. However, coverage remains limited-especially in the lower-EXT and lower-EST regions-indicating insufficient representation of diverse psychological profiles. By contrast, our **Ours (Resample)** method (top left) yields outputs that closely match the human distribution, covering the full trait space and successfully reaching underrepresented regions. This strongly supports our quantitative findings in Section 4.1, affirming the effectiveness of our two-stage sampling approach. Although our method achieves substantial improvements in trait alignment, the generated distribution still falls short of fully matching real human data, due to inherent model and data biases. Closing this gap remains a key challenge.

4.4 Group-Specific Persona Adaptation

We assess whether the globally aligned pool $\mathcal{P}'$ can be adapted to specific demographic targets using two test sets: YRBSS [3] and WVS [25]. For each group, we issue a natural-language query (e.g., “Generate a persona from the Hong Kong region”) and retrieve the top- K personas from $\mathcal{P}'$ using an embedding model. Specifically, we use **Qwen3-Embedding-0.6B** [83], which maps both queries and personas into a shared semantic space. The corresponding training hyper-parameters and prompt details for the embedding model can be found in Appendix B.4, B.5 and B.6. Retrieved personas are then revised using Qwen2.5-72B-Instruct to produce 1,000 group-specific personas per setting. We compare three variants: (1) **Resample** (from Section 3.2); (2) **EM w/o train**, which retrieves from Resample using an *untrained* embedding model; and (3) **EM w train**, our full method with contrastive training. Alignment is evaluated using AMW, FD, SW, and MED, averaged by region. As shown in Table 4,

EM w train achieves the lowest errors across all regions and metrics, reducing overall alignment error by 19.1% versus AlignX and 8.9% versus Resample. Even **EM w/o train** outperforms all external sets, validating the effectiveness of the retrieval approach. A comparison between EM w/o train and EM w train reveals consistent improvements across all baselines and metrics, highlighting that the proposed contrastive training enhances generalization rather than overfitting to a single domain.

5 Conclusion

We propose a principled framework for constructing population-aligned persona sets for LLM-based social simulation. Existing approaches often focus on individual-level attributes while neglecting population-level distributional fidelity, leading to biased and unrealistic outcomes. Our three-stage pipeline (1) extracts high-quality personas from real-world social media, (2) performs two-stage alignment via importance sampling and optimal transport, and (3) enables group-specific persona generation. Across six different psychometric tests, our method consistently outperforms widely used persona sets in both population alignment and behavioral realism. It generalizes well across psychometric instruments and supports high-fidelity group-specific simulation. These results highlight the importance of distribution-aware persona construction for reliable social modeling. Future work will explore high-order simulations for forecasting social trends and evaluating intervention strategies.

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A Notation

Table 5 and Table 6 summarize the notations used in our paper. Part I corresponds to Section 3.1. Part II and Part III correspond to Stage 1 and Stage 2 of Section 3.2, respectively. Part IV corresponds to Section 3.3.

Table 5: Summary of notation (A): Seed mining, LLM responses, and KDE-based Importance Sampling.

Symbol	Description
Part I: Basic objects & LLM-induced responses	
$\mathcal{P} = \{p_i\}_{i=1}^N$	Set of textual personas (Sec. 3.1)
N	Number of personas in $\mathcal{P}$
s	Aggregated long-form user text (concatenated posts)
$\mathcal{G}$	LLM persona summarizer mapping $s \mapsto p_i$
$p_i = \mathcal{G}(s)$	Narrative persona synthesized from s
$Q = \{q_k\}_{k=1}^d$	Psychometric items (e.g., IPIP Big Five)
d	Number of items / response dimension
LLM	Frozen LLM mapping (p_i, q_k) to a scalar response
$\parallel$	Textual concatenation operator
x_{ik}	Model response to (p_i, q_k)
$x_i \in \mathbb{R}^d$	Response vector for persona p_i
$X \in \mathbb{R}^{N \times d}$	Persona response matrix with rows x_i
$Y = \{y_j\}_{j=1}^M$	Human response vectors from surveys
M	Number of human response samples
r_{persona}	Empirical distribution of $\{x_i\}_{i=1}^N$
r_{human}	Empirical distribution of $\{y_j\}_{j=1}^M$
Part II: KDE-based Importance Sampling	
h	Bandwidth parameter for KDE
I	Identity matrix
$\mathcal{N}(z; h^2 I)$	Multivariate Gaussian kernel density
$\hat{r}_{\text{persona}}$	KDE estimate of r_{persona}
$\hat{r}_{\text{human}}$	KDE estimate of r_{human}
w_i^{IS}	IS weight: $\hat{r}_{\text{human}}(x_i) / \hat{r}_{\text{persona}}(x_i)$
π_i	IS sampling probability for p_i (normalized w_i^{IS})
$N^\dagger$	Number of candidates sampled by IS
$\mathcal{P}^\dagger$	Candidate persona subset after IS
$X^\dagger$	Response vectors of $\mathcal{P}^\dagger$

B Further Implementation Details

B.1 Experimental Setup

All experiments were conducted on a high-performance computing server equipped with eight NVIDIA A100 GPUs, each with 40 GB of memory. The models referenced in the main text correspond to specific checkpoints available on HuggingFace: for Llama-3-70B, we use meta-llama/Meta-Llama-3-70B-Instruct²; for Qwen2.5-72B, we use Qwen/Qwen2.5-72B-Instruct³; and for Phi-4, we use microsoft/phi-4⁴. Our seed persona mining pipeline draws from three open-source datasets. The blog data is based on the Blog

²<https://huggingface.co/meta-llama/Meta-Llama-3-70B-Instruct>

³<https://huggingface.co/Qwen/Qwen2.5-72B-Instruct>

⁴<https://huggingface.co/microsoft/phi-4>

Table 6: Summary of notation (B): Entropic OT alignment and Group-specific construction.

Symbol	Description
Part III: Optimal Transport alignment	
ω_k	Item weight for prompt q_k
$C \in \mathbb{R}^{N^\dagger \times M}$	Cost matrix, $C_{ij} = \sum_k \omega_k (x_{ik}^\dagger - y_{jk})^2$
ε	Entropic regularization parameter in OT
$\mathbf{a}^\dagger, \mathbf{b}$	Uniform marginals on candidates / human samples
$\mathbf{1}_n$	All-ones vector of length n
$\Pi(\mathbf{a}^\dagger, \mathbf{b})$	Feasible couplings with given marginals
$\langle C, \Gamma \rangle$	Frobenius inner product $\sum_{ij} C_{ij} \Gamma_{ij}$
K	Gibbs kernel: $K_{ij} = \exp(-C_{ij}/\varepsilon)$
u, v	Sinkhorn scaling vectors
$\Gamma, \Gamma^\star$	Transport plan / optimal plan
$\text{diag}(\cdot)$	Diagonal matrix formed from a vector
w_i^{OT}	OT-derived weight for $x_i^\dagger$
π_i^{OT}	Sampling probability induced by w_i^{OT}
N'	Final number of personas sampled after OT
$\mathcal{P}'$	Final population-aligned persona subset
$W_2(\cdot, \cdot)$	2-Wasserstein distance
Part IV: Group-specific retrieval & generation	
q	Natural-language group/query
$\text{Embed}(\cdot)$	Learned embedding model for queries/personas
e_q, e_i, e_p	Embeddings of q , persona p_i , and a generic persona
$\text{sim}(\cdot, \cdot)$	Cosine similarity between embeddings
K_{ret}	Top- K retrieval size (distinct from OT kernel K)
$\mathcal{P}_{\text{seed}}(q)$	Retrieved seed personas for q
$\mathcal{P}_{\text{group}}(q)$	Generated group-specific personas conditioned on seeds
e_p^+	Positive persona embedding in contrastive loss
p^-	Negative personas in contrastive loss
$\mathcal{L}$	Contrastive loss: $-\log \frac{e^{\text{sim}(e_q, e_p^+)}}{e^{\text{sim}(e_q, e_p^+)} + \sum_{p^-} e^{\text{sim}(e_q, e_p^-)}}$

Authorship Corpus⁵, which includes entries from over 30000 users. For population-level persona sampling via Importance Sampling and Optimal Transport (Section 3.2), we apply a Gaussian KDE with bandwidth 0.20 and retain the 70% of candidate personas with the highest importance weights. OT weights are computed in batches of size 10,000 with entropic regularization $\varepsilon = 0.08$ (scaled to the median cost). The Sinkhorn algorithm is run for 250 iterations. Finally, for embedding model training, we follow the SWIFT recipe from the Qwen3-Embedding repository, available at this link⁶.

B.2 Datasets

PIF Big Five psychometric test [26]. The Big Five Inventory consists of 50 self-descriptive statements (e.g., "I see myself as someone who is talkative") rated on a 5-point Likert scale. It quantifies five broad personality dimensions-Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism-providing stable trait

⁵https://huggingface.co/datasets/barilan/blog_authorship_corpus

⁶<https://github.com/QwenLM/Qwen3-Embedding/blob/main/docs/training/SWIFT.md>

profiles. The instrument is used to predict outcomes in areas such as job performance, relationship quality, and health behaviors, and to examine correlations between personality and cognitive, emotional, or social variables.

CFCS [60]. The Consideration of Future Consequences Scale (CFCS) is a 12-item self-report instrument rated on a 5-point Likert scale designed to quantify individual differences in the extent to which people consider distant versus immediate outcomes of their actions. The CFCS score predicts a wide range of real-world behaviors, such as health maintenance, financial planning, environmental stewardship, and academic persistence, making it a versatile benchmark for assessing temporal orientation and decision-making dynamics in both clinical and applied research contexts.

FBPS [48]. The Firstborn Personality Scale is a 25-item measure designed to capture personality traits statistically associated with being firstborn, without explicitly referencing birth order. The items reflect a profile marked by intellectual engagement, verbal interests, competitiveness, assertiveness, and cognitive intensity-traits such as enjoying complex reading, using advanced vocabulary, following politics, and striving to outdo others. Developed through large-sample analysis, the FBPS offers a concise snapshot of a driven, literate, and analytically minded personality style.

Duckworth. [14]. The Duckworth Grit Scale assesses perseverance and passion for long-term goals through 12 self-report items (e.g., "I finish whatever I begin," "Setbacks don't discourage me"), providing a validated measure of sustained effort and consistency of interest. In this paper, Duckworth serves as a self-regulation benchmark to verify that our persona alignment method captures realistic distributions of grit and motivational persistence.

WVS [25]. The World Values Survey administers 200-300 questions on political attitudes, social norms, religious beliefs, and subjective well-being (e.g., "Overall, how happy are you?") across over 100 countries every 5-10 years. It tracks longitudinal changes in values related to democracy, gender roles, economic security, and quality of life. WVS supports comparative research in sociology, political science, and international development.

YRBSS [3]. Conducted by the CDC, YRBSS collects data on health-risk behaviors among U.S. high school students. Topics include tobacco/alcohol/drug use, sexual activity, mental health (e.g., suicidality), nutrition, physical activity, and violence exposure. The goal is to monitor trends and inform public health interventions targeting adolescents.

B.3 Dataset Statistics

Table 7 summarizes all datasets used in our study. For the World Values Survey (WVS), we construct three regional subsets to evaluate the Group-Specific Persona Adaptation module, as described in Section 4.4. These subsets correspond to the full set of countries available within each target region in the WVS dataset: East Asia, including China, Japan, South Korea, Hong Kong, Mongolia, Taiwan, and Macao; North America, including Canada, United States, and Mexico; and Europe, including Germany, Netherlands, and United Kingdom.

Table 7: Statistics of psychometric and behavioral survey datasets used for alignment and evaluation.

Dataset	# Questions	# Human Responses	# Countries and Regions
Big Five (IPIP)	50	1,015,341	224
CFCS	12	15,034	144
FBPS	76	41,841	169
Duckworth	50	4,270	111
YRBSS	105	20,103	1
WVS (Full)	294	97,221	66
WVS – East Asia	250	11,593	7
WVS – North America	250	8,355	3
WVS – Europe	250	6,282	3

B.4 Query Generation from Personas for Embedding Training

Motivation. To construct positive (*query*, *persona*) pairs for the contrastive encoder (§3.3), we automatically derive two natural language queries from each persona $p_i \in \mathcal{P}$. The first is a *detailed* query that captures fine-grained persona traits, and the second is a *broad* query that summarizes only high-level characteristics. We generate both queries using LLM prompting, with one prompt designed for detailed descriptions (see Prompt 1) and another for coarse-grained abstractions (see Prompt 2). Each query is paired with the original persona for training the embedding model using the contrastive objective defined in Eq. 15.

Listing 1: Prompt for detailed (long) query generation

You are a helpful assistant tasked with generating queries based on detailed persona descriptions.

Your goal is to generate concise queries that describe the key traits of the persona, such as age, gender, emotional tone, and significant interests or lifestyle. The query should be specific but short, capturing the persona's core characteristics without going into excessive detail.

Persona: "The user is a young, likely female, living in a town she affectionately calls 'peach town', working multiple jobs, including one at a gym and another at a cleaners, and values relaxation and enjoyment of simple pleasures, like good food and sunny days, while navigating life's changes and emotional ups and downs with an introspective and slightly moody tone."

Generated query: "Generate a young female persona who works multiple jobs, values relaxation and simple pleasures, and has an introspective and slightly moody tone."

Your generated query should capture the persona's essential traits in a clear and concise manner, avoiding unnecessary complexity.

Only return the generated query, nothing else.

Based on the following persona description, generate a concise query that describes a persona with similar traits

and characteristics. Focus on the persona's key traits like age, gender, emotional tone, and major activities or interests, keeping the query short and to the point.

Persona description:
{persona}

Generated query:
Only return the generated query, nothing else.

Listing 2: Prompt for broad (short) query generation

You are a helpful assistant tasked with generating broader queries based on persona descriptions. Your goal is to generate simple and concise queries that describe the basic traits of the persona, such as age, gender, and general interests or activities. The query should be high-level, focusing on the core attributes, and should be short without diving into personal or emotional details.

Persona: "The user is a young, likely female, living in a town she affectionately calls 'peach town', working multiple jobs, including one at a gym and another at a cleaners, and values relaxation and enjoyment of simple pleasures, like good food and sunny days, while navigating life's changes and emotional ups and downs with an introspective and slightly moody tone."
Generated query: "Generate a young female persona who works multiple jobs."

Your generated query should focus on broad characteristics and general interests, and should be brief without over-explaining.

Only return the generated query, nothing else.

Based on the following persona description, generate a broad, concise query that describes a persona with similar traits and characteristics. Focus on the basic attributes such as age, gender, and major interests or lifestyle, keeping the query short and general.

Persona description:
{persona}

Generated query:
Only return the generated query, nothing else.

Training usage. For each persona we generate one detailed query with Prompt 1 and one broad query with Prompt 2. Both are encoded and paired with the same persona embedding in the contrastive loss of Eq. (15), while hard negatives are mined as described in the main text. Full tuning hyper-parameter details are provided in Table 8.

B.5 LLM-based Filtering of Semantically Matched Negatives

To improve the quality of negative sampling during training (§3.3), we apply an additional filtering step to eliminate false negatives personas that are semantically aligned with the query but incorrectly

Table 8: Contrastive embedding model fine-tuning hyper-parameters.

Hyper-parameter	Assignment
Number of epochs	2
Batch size per device	4
Gradient accumulation	4
Effective batch size	32
Learning rate	6e-6
Precision	FP16
Optimizer	Deepspeed (ZeRO-3)

treated as negatives. Specifically, for each query q_i , we examine two sets of candidate negatives: the top- N most similar personas (in embedding space) and N randomly sampled personas. Both sets may contain cases that match the query, so we prompt a large language model to verify alignment. Given a query-persona pair $(q_i, \tilde{p})$, the LLM is asked to return a strict binary response: YES if the persona clearly matches the query, and NO otherwise. Only those pairs labeled NO are retained as valid negatives. The prompt template used for this filtering step is shown in Prompt 3.

Listing 3: Prompt for LLM-based semantic match filtering

You are an expert evaluator tasked with determining whether a user persona matches a given query or description.

Your job is to carefully analyze if the persona accurately represents someone who would be described by the query. Consider:

Key characteristics mentioned in the query

Interests, behaviors, and preferences

Demographics if mentioned

Overall alignment between query and persona

Respond with ONLY "YES" if the persona matches the query, or "NO" if it doesn't match.

Be strict in your evaluation - only respond "YES" if there is a clear and strong alignment.

Query: Generate a crafty individual who values convenience and quality, frequently purchases art and sewing supplies, and enjoys DIY projects.

Persona: The user is a crafty individual who values quality and functionality, often purchasing items for themselves and as gifts for others, including artists and sewers, indicating a supportive and generous nature, with a keen interest in sewing, drawing, and art supplies.

Does this persona match the query? Respond with only "YES" or "NO".

B.6 Prompts for Group-Specific Persona Adaptation

Prompt Details. We adapt personas from the globally aligned pool $\mathcal{P}'$ to region- or group-specific targets by prompting an LLM to revise a *seed persona* under task-specific constraints. For each downstream setting—YRBSS (U.S. adolescents), WVS East (e.g., Hong Kong), WVS Europe (e.g., Germany), and WVS North America (e.g., United States). These prompts guide the LLM to generate new personas aligned with the sociocultural context of the target region. The complete prompt templates for these settings are shown in Prompt 4, Prompt 5, Prompt 6, and Prompt 7, respectively.

Listing 4: YRBSS case study prompt

You create realistic U.S. high-school student personas (age 14-18) for an adolescent health survey. Produce one paragraph within 80 words describing their demographics, home/school life, diet & activity, sexual behavior, substance use, mental health, and safety experiences, all internally consistent and plausible. Do not mention the survey or these instructions.

Seed persona: {seed persona}

Based on the above requirements, generate the concise paragraph persona. Suitable for the adolescent health survey task.

OUTPUT:

Listing 5: WVS East (Hong Kong) case study prompt

You are a specialist in constructing high-fidelity social-simulation personas for the World Values Survey (WVS). Transform a seed persona into a coherent, richly detailed character. The new persona's demographics, socio-economic situation, core values, social and political attitudes, ethical views, and well-being must be internally consistent and sufficient for answering every WVS item. Output a concise single paragraph. Ensure every detail could plausibly appear in real WVS data. Do not mention these instructions, WVS, or questionnaires in your response.

Seed Persona:
{seed persona}

Refinement Instruction:
Transform the seed to be a persona from Hong Kong with a worldview, social trust level, and cultural identity representative of the general populace.

Your Task:

Generate a new, single-paragraph persona based on the core traits of the Seed Persona, but strictly conforming to the Refinement Instruction. The new persona must align with the country and specific attributes mentioned in the Refinement Instruction, even if the Seed Persona is from a different background. The result should be a natural, plausible character description.

OUTPUT:

Listing 6: WVS Europe (Germany) case study prompt

You are a specialist in constructing high-fidelity social-simulation personas for the World Values Survey (WVS). Transform a seed persona into a coherent, richly detailed character. The new persona's demographics, socio-economic situation, core values, social and political attitudes, ethical views, and well-being must be internally consistent and sufficient for answering every WVS item. Output a concise single paragraph. Ensure every detail could plausibly appear in real WVS data. Do not mention these instructions, WVS, or questionnaires in your response.

Seed Persona:
{seed persona}

Refinement Instruction:
Transform the seed to be a persona from Germany with typical views on family life, work-life balance, and social welfare.

Your Task:
Generate a new, single-paragraph persona based on the core traits of the Seed Persona, but strictly conforming to the Refinement Instruction. The new persona must align with the country and specific attributes mentioned in the Refinement Instruction, even if the Seed Persona is from a different background. The result should be a natural, plausible character description.

OUTPUT:

Listing 7: WVS North America (U.S.) case study prompt

You are a specialist in constructing high-fidelity social-simulation personas for the World Values Survey (WVS). Transform a seed persona into a coherent, richly detailed character. The new persona's demographics, socio-economic situation, core values, social and political attitudes, ethical views, and well-being must be internally consistent and sufficient for answering every WVS item. Output a concise single paragraph. Ensure every detail could plausibly appear in real WVS data. Do not mention these instructions, WVS, or questionnaires in your response.

Seed Persona:
{seed persona}

Refinement Instruction:

Transform the seed to be a persona from the United States with mainstream perspectives on national identity, civic engagement, and social justice.

Your Task:

Generate a new, single-paragraph persona based on the core traits of the Seed Persona, but strictly conforming to the Refinement Instruction. The new persona must align with the country and specific attributes mentioned in the Refinement Instruction, even if the Seed Persona is from a different background. The result should be a natural, plausible character description.

OUTPUT:

Case Studies. We illustrate the group-specific persona construction process through four case studies, each comprising a raw seed persona and its LLM-generated output persona after adaptation to the target setting. These outputs follow the prompts detailed in Table 9.

C Privacy De-identification

Before Stage 2, we insert a lightweight *de-identification* step to ensure that no sensitive personal information (PII) is exposed when personas are inspected or released. For every persona in the OURS (RAW DATA) reported in Table 1, we first run a GPT-4o PII detector. The detector prompt is shown in Listing 8. The model considers as sensitive any full name, street-level home address, email address, phone number, or bank/credit-card number. Across the corpus, 99.31% of personas are flagged no, indicating that no PII is present. Listing 9 provides an example that triggers a yes response, while Listing 10 illustrates a typical no. For every record flagged yes, we invoke a second prompt (Listing 11) that rewrites the persona by replacing only the sensitive fields with realistic but fictitious values while preserving style and meaning. A de-identified output corresponding to the yes case is presented in Listing 12. We further conducted a human study on all these rewritten yes responses, involving five independent annotators. All cases were verified to have the identified PII correctly removed or substituted, with no observed semantic inconsistencies, indicating the effectiveness of the de-identification process.

Listing 8: PII detection prompt

You are tasked with checking persona text for sensitive personal information.

Sensitive items are defined as follows:

- Full names (at least first and last name together, not just a first name or a generic reference).
- Home addresses (street-level detail, not just city or country).
- Email addresses (must contain '@' with a valid-looking format).
- Phone numbers (must contain at least 7 digits, optionally with separators like '-', '()', or spaces).
- Bank account or credit card numbers (must contain a sequence of at least 8 digits, e.g., full or partial account numbers).

Do NOT treat the following as sensitive:

- Generic mentions of 'phone number', 'email', 'credit card', 'parents', etc. without specific values.

- General locations (city, country, region).
- Public figure names (e.g., celebrities, politicians, authors).

Instructions:

1. If you detect any concrete sensitive information as defined above, output only: yes.

2. If none is found, output only: no.

Output must be a single word: yes or no.

Persona text:

Listing 9: Detector output: yes

The user is a young, likely married, British mother in her mid-to-late twenties, living with her husband, Mr. Spid, and their young daughter, Baby Spid. She values her relationships and family, but also enjoys her independence, music, and socializing with friends, often expressing herself through sarcastic humor and frustration

Listing 10: Detector output: no

The user is a young male, likely in a relationship, who values simplicity and frugality, but also enjoys spending time with his girlfriend and having fun, often finding joy in everyday moments, with a carefree and lighthearted tone, living in the Lower Mainland, possibly in Canada, given the mention of T&T and Tim Horton's.

Listing 11: PII rewrite prompt

You are tasked with de-identifying the given persona text by replacing sensitive items with realistic but fictitious substitutes.

Sensitive items to REPLACE: full names, home addresses, email addresses, phone numbers, and bank account numbers.

Do NOT replace nationality or general location (city, country, region).

Requirements:

- + Preserve the original style, tone, and intent.
- + Do not leak or partially reveal the original sensitive values.
- + Generate substitutes that look plausible but are fabricated and not traceable to real individuals.

Listing 12: Rewritten persona (de-identified)

The user is a young, likely married, British mother in her mid-to-late twenties, living with her husband, Tom Ridley, and their young daughter, Lila Ridley. She values her relationships and family, but also enjoys her independence, music, and socializing with friends, often expressing herself through sarcastic humor and frustration.

Table 9: Examples of seed personas and their LLM-generated outputs across different demographic adaptation tasks. Each output persona reflects the cultural, behavioral, and social context of the respective group.

Setting	Seed Persona	Output Persona
YRBSS (U.S. teens)	The user is a young adult, likely female, living in Pen-ticton, Canada, who values a healthy lifestyle and strives to maintain an active routine, often starting her day with yoga, walks, or other exercises, despite facing challenges with energy and motivation, particularly due to her early work schedule and occasional injuries.	Alex, 16, lives in a suburban area of California with her parents and younger brother. She is a junior at a public high school, where she excels in academics and is on the track team. Alex starts her day with a 30-minute jog and eats a balanced diet. She has had minor injuries but stays committed to fitness. She is a virgin, avoids substances, and manages stress through journaling and talking to friends.
WVS East (Hong Kong)	The user is a young, likely Asian, possibly Chinese, individual who values cultural heritage and identity, is emotional and sentimental, and is openly gay and appreciative of diversity.	Alex Wong, 28, lives in Hong Kong and works as a graphic designer. He embraces Chinese heritage in his work, is openly gay, and actively supports LGBTQ+ causes. He visits ancestral sites for emotional grounding, participates in community service, and supports democratic reform. He balances sentimentality with a pragmatic mindset.
WVS Europe (Germany)	The user is a 25-year-old married male, living with his wife, earning \$145,000–\$160,000 jointly, focused on saving for a house, and values work-life balance.	Markus, 25, lives in Munich with his wife Lisa. Both work in tech and earn €130,000–€145,000 annually. Markus prioritizes flexible hours and vacation time. They’re saving for a home and planning for children. He supports welfare policies and environmental volunteering, favoring collective well-being over individualism.
WVS North America (U.S.)	The user is a likely young, liberal-minded individual, possibly American, who values critical thinking and challenges norms, with strong views on justice and social responsibility.	Emily Thompson, 32, lives in suburban Chicago. A marketing professional and moderate Democrat, she volunteers locally, values civic duty, and supports social justice. She votes regularly, discusses reform topics, and values honesty and responsibility. She is optimistic despite student debt and owns two rescue dogs.

D Metric Definitions

In this appendix, we provide formal definitions of the evaluation metrics used in Section 4, covering both population-level alignment and individual-level consistency.

Population-level alignment. We compare the distribution of LLM-generated responses with real human responses using four standard metrics:

Averaged Monotonic Wasserstein (AMW) [65, 66] measures 1D Wasserstein-1 distance across each psychometric trait and averages over all N traits:

$$\text{AMW} = \frac{1}{N} \sum_{t=1}^N \int_{-\infty}^{\infty} |F_{P_t}(x) - F_{Q_t}(x)| dx, \quad (17)$$

where F_{P_t} , F_{Q_t} are empirical CDFs of human and synthetic responses on trait t .

Fréchet Distance (FD) [28, 64] compares two multivariate Gaussian distributions:

$$\text{FD} = \|\mu_P - \mu_Q\|_2^2 + \text{Tr}(\Sigma_P + \Sigma_Q - 2(\Sigma_P \Sigma_Q)^{1/2}), \quad (18)$$

where μ_P, μ_Q are mean vectors and Σ_P, Σ_Q are covariance matrices of human and synthetic samples.

Sliced Wasserstein Distance (SW) [40, 45] projects high-dimensional samples onto K random directions v_i , then averages 1D Wasserstein distances:

$$\text{SW} = \frac{1}{K} \sum_{i=1}^K W_1(\langle X, v_i \rangle, \langle Y, v_i \rangle), \quad (19)$$

where $\langle \cdot, v_i \rangle$ denotes projection.

Maximum Mean Discrepancy (MMD) [30, 52] measures distributional gap in a reproducing kernel Hilbert space (RKHS):

$$\text{MMD} = \|\mathbb{E}_{x \sim P}[\phi(x)] - \mathbb{E}_{y \sim Q}[\phi(y)]\|_{\mathcal{H}}, \quad (20)$$

where ϕ is the kernel mapping and $\mathcal{H}$ is the RKHS.

Individual-level consistency. To evaluate whether inter-trait correlations are preserved, we compute Pearson correlation coefficients across all N trait pairs from both human and synthetic responses, and report their mean absolute error:

$$\text{MAE}_{\text{corr}} = \frac{1}{N} \sum_{i=1}^N |\hat{\rho}_i - \rho_i^{\text{human}}|, \quad (21)$$

where ρ_i^{human} and $\hat{\rho}_i$ denote the Pearson correlation coefficient [1, 51] for the i -th trait pair in human and synthetic data, respectively.

E Limitations

Our filtering pipeline removes harmful content such as violence, harassment, and privacy violations to ensure persona safety and ethical compliance. However, this may introduce a mild positivity bias, as expressions involving personal adversity or social tension may be underrepresented. This reflects a common trade-off between safety and realism in LLM-based social simulation. Moreover, both our seed corpora and the reference psychometric datasets used for alignment (e.g., the IPIP Big Five inventory [26]) are derived from internet-accessible populations. As a result, individuals with limited or no online presence, such as those in rural, elderly, or under-resourced communities, may be underrepresented. It is also an inherent challenge in large-scale population modeling using publicly available digital traces.

F Seed Persona Mining: Data and Prompts

Corpora and Scale. We construct the seed persona pool from a large-scale, human-authored corpora that reflect personal narratives and behavioral cues. Specifically, we use the *Blog Authorship Corpus*, which contains 681K posts and roughly 140 million words from over 19,000 bloggers; Collectively, these corpora capture diverse first-person perspectives spanning various demographics, interests, and socioemotional expressions.

Preprocessing. We design modality-specific filters to remove noise while preserving personal expression. For blogs, we remove segments shorter than 30 tokens or lacking first-person pronouns, which tend to be non-narrative. The remaining texts are passed to *Llama-3.3-70B-Instruct* with a specialized cleaning prompt (Prompt 13) that removes HTML tags, URLs, non-English snippets, mentions, and other noise while preserving tone and syntax. This step yields approximately 500K blog entries.

Raw Data Quality Control. Each cleaned text is further annotated using *Llama-3.3-70B-Instruct* via Prompt 14, which assigns: (1) a three-level <quality> tag—*high*, *medium*, or *low*, reflecting narrative depth and relevance to human thought or behavior; and (2) a binary <harmless> tag indicating whether the content avoids sensitive or unethical material (e.g., hate speech, violence, privacy violations). Only texts labeled both high quality and harmless are retained.

Persona Generation. For each retained user, we aggregate their posts and remove authors with fewer than six entries. The remaining user-level documents are concatenated into a single segment s and passed to *Llama-3.3-70B-Instruct* using Prompt 15, which extracts a narrative persona paragraph. The persona is written in third person and integrates salient demographic, psychological, and behavioral traits, but only if they are explicitly supported by the source text. If the input fails to support persona synthesis, the model outputs NULL to avoid hallucination.

Persona Quality Control. Each generated persona is scored by a critic LLM using the evaluation rubric in Prompt 16. Five dimensions are rated on a 1-10 scale: *hallucination* (factual grounding), *coverage* (extent of salient information captured), *conciseness* and *clarity*, *relevance* (exclusion of generic/promotional content), and an *overall*

score. We retain only those personas where the overall score exceeds 8, and each sub-score exceeds 7. This step filters out 27.8% from blogs.

Final Seed Pool. The resulting seed persona pool comprises over 100,000 high-fidelity, diverse narrative personas, spanning a broad spectrum of psychometric and demographic dimensions. These personas serve as the foundation for downstream alignment and targeted generation modules.

Prompts Detail. To ensure reproducibility, we include the full text of all LLM prompts used during the pipeline—cleaning (Prompt 13), quality control (Prompt 14), persona generation (Prompt 15), and persona evaluation (Prompt 16).

Listing 13: Prompt A: rewrite

```
You are a text cleaning assistant.
Your task is to clean a blog post by removing noisy or
irrelevant elements while fully preserving the original
content's meaning, sentence structure, punctuation,
and tone.

Apply the following cleaning rules:
1. Remove truly extraneous characters and encoded text,
   including:
   - Unicode symbols like \u00a3, \u2026, \u00ca\u00cc, etc
   - HTML entities like &;, >;, #x200B, etc.
   - Repetitive filler patterns like . . ., \n.\n.\n, or
     similar.
2. Delete all URLs, including both raw links (https://...,
   http://...) and markdown-style links ([text](url)).
3. Remove all hashtags (e.g., #LifeGoals, #AI2024) unless
   they are clearly essential to the post's meaning.
4. Remove all user mentions (e.g., @elonmusk, @user123). If
   the mention is critical for understanding or its
   removal breaks the sentence structure, substitute an
   appropriate placeholder (e.g., "a friend," "the author
   ") based on context.
5. Strip out emojis, meme-style text, ASCII art, and
   decorative symbols that do not contribute meaning.
6. Discard non-English text or phrases only if they are not
   integrated into natural English usage. Common loanwords
   or cultural terms (e.g., "sushi," "futbol") should be
   preserved if they are relevant.

Important Guidelines:
- Do not remove or alter meaningful punctuation such as
  parentheses (), dollar signs $, question marks ?, or
  exclamation marks ! if used naturally.
- Do not rephrase, paraphrase, or rewrite any part of the
  original.
- Maintain all meaningful sentence boundaries and user
  expression styles.
- If a removal creates broken syntax or unclear meaning,
  minimally repair the sentence to ensure readability
  without altering the author's voice.
- Output the cleaned text only-no explanations or notes.

Here is the blog post to clean:
{blog}
```

Listing 14: Prompt B: raw data quality control

You are a helpful assistant that helps me determine the data quality of a blog. The background is that I want to collect blogs which contain human behaviors or human thoughts, so that I can further study social science based on the collected data in the next step. However, as you may know, blogs from the Internet contain various types of content, and many of them are irrelevant to my goal, so I need to filter them out.

Typically, a blog's quality is high if it records detailed events of a human, reflects human life, mentions social behaviors, or reveals the author's thoughts or feelings about something.

A blog's quality is medium if it only briefly mentions some content related to human behaviors or thoughts, but lacks enough context to understand a complete story or perspective.

A blog's quality is low if it has nothing to do with human behaviors or thoughts, such as ads, job posts, company descriptions, fictional plots, random word dumps, and other irrelevant types. Additionally, a blog is also low quality if it is filled with non-English words, URLs, mentions (e.g., @user), hashtags, special characters (such as Unicode symbols or HTML entities), or emojis, which suggest noise or lack of meaningful personal expression.

In addition to tagging the quality, please also determine whether the blog is harmless. A blog is considered harmless if it does not contain unethical or sensitive content such as violence, pornography, or privacy violations. If such content exists, the harmless tag should be no; otherwise, it should be yes.

So your task is to tag the blog in two aspects:

A quality tag, which can be either "high", "medium", or "low".

A harmless tag, which can be either "yes" or "no".

Please output both tags in the following XML format, and do not include any other words or explanations:

```
<output><quality>...</quality><harmless>...</harmless></output>
```

Below is the user's blog post:
{blog}

Listing 15: Prompt C: persona_generation

Analyze the provided blog posts to create a concise and vivid user persona in a single cohesive paragraph (no more than 100 words). Seamlessly weave together important aspects of the user's persona, such as demographics (age, gender, profession, nationality, location, marital status), key personality traits, core values, interests, and emotional tone, only when these aspects are explicitly supported by the raw blogs. Extract persona based on the user's salient events or thoughts.

Exclude generic, promotional, or repetitive content in the raw blogs (e.g., product ads, event schedules, technical knowledge re-post).

Write in third-person (e.g., 'The user is a...'), avoiding lists or bullet points. If the blogs' content cannot support high-quality persona extraction, simply output "NULL".

Now the task begins. Below are the person's raw blogs. Please only output the result. Do not include any additional explanatory text.

{blog}

Listing 16: Prompt D: persona_quality_control

****Evaluate the quality of a generated user persona based on the raw posts and the extracted persona paragraph provided. Consider the following aspects in your evaluation:****

- **Hallucination:**** Does the persona strictly reflect information that is explicitly supported by the raw posts? Score 10 for no hallucinations (all facts directly mentioned), 8 if a few facts are very confidently inferred, down to 1 if most facts are neither mentioned nor plausibly inferred.
- **Coverage:**** Does the persona cover the main salient aspects of the user that appear in the raw posts (e.g., demographics, values, interests, emotional tone)? Are important elements missing?
- **Conciseness and Clarity:**** Is the persona concise, cohesive, and written in clear third-person prose within the 100-word limit? Does it avoid redundancy, listing, and repetition?
- **Relevance:**** Has generic, promotional, or repetitive content from the raw posts been appropriately excluded from the persona?
- **Overall:**** Give an overall score for the persona based on the above aspects and your holistic judgment.

****Output your evaluation in the following XML-like format:****

```
<data>
  <explanation> [Your brief analysis and justification.] </explanation>
  <hallucination> [Hallucination score: integer 1-10] </hallucination>
  <coverage> [Coverage score: integer 1-10] </coverage>
  <conciseness> [Conciseness and clarity score: integer 1-10] </conciseness>
  <relevance> [Relevance score: integer 1-10] </relevance>
```

```

<overall> [Overall score: integer 1-10] </overall>
</data>

**Instructions:**
Base all ratings and analysis strictly on the given raw
posts and generated persona. Do not include any content
or facts unsupported by the original posts. If the
persona reports "NULL", give the 1 to all scores.

Now the task begins.

```

Data Statistics. To provide a quantitative overview of the entire seed persona construction pipeline, Table 10 summarizes the data retained at each major processing stage. The column labels in the table use abbreviated names that correspond directly to the sections described above: “Raw” refers to the original crawled texts (**Corpora and Scale**); “Preproc” denotes the number of texts remaining after preprocessing and cleaning (**Preprocessing**); “Raw QC” corresponds to texts retained after quality and safety filtering (**Raw Data Quality Control**); “Persona Gen” indicates the number of users for whom a draft persona was generated (**Persona Generation**); and “Persona QC” reflects the final number of high-quality personas retained after evaluation (**Persona Quality Control**). Statistics are shown separately for blogs.

G Theoretical Guarantees

In this part, we provide finite-sample guarantees for our two-stage persona alignment framework. We quantify the approximation error after Importance Sampling and the additional bias introduced by entropic Optimal Transport. These results provide formal support for the population-level alignment effectiveness of our method.

Theorem 1 (Approximation Error after Importance Sampling). Let $\mathcal{P}^\dagger = \{p_i\}_{i=1}^{N^\dagger}$ be the set of candidate personas selected via importance sampling from an original set of N personas, and let r_{human} denote the empirical human response distribution from M samples. Let $\hat{r}_{\text{persona}}^\dagger$ be the empirical distribution of the responses induced by $\mathcal{P}^\dagger$, and d be the dimension of response vectors. Then with probability at least $1 - \delta$, we have:

$$W_2(\hat{r}_{\text{persona}}^\dagger, r_{\text{human}}) = O\left(h^2 + \sqrt{\frac{\log(1/\delta)}{Nh^d}} + \sqrt{\frac{\log(1/\delta)}{Mh^d}} + \sqrt{\frac{\log(1/\delta)}{N^\dagger}}\right) \quad (22)$$

where $h > 0$ denotes the kernel bandwidth for KDE. This result confirms that the candidate personas selected through KDE-based importance sampling already achieve meaningful coarse-grained alignment with human psychometric distributions.

Theorem 2 (Entropic OT Approximation Error). Let $X^\dagger = \{x_i\}_{i=1}^{N^\dagger}$ be the response vectors of personas in $\mathcal{P}^\dagger$, and $Y = \{y_j\}_{j=1}^M$ be the human responses. Let $C \in \mathbb{R}^{N^\dagger \times M}$ be the cost matrix with entries $C_{ij} = \|x_i - y_j\|_2^2$, and let $\Gamma^\star$ be the optimal transport plan obtained via entropic OT with regularization parameter $\varepsilon > 0$. Then the deviation from exact Wasserstein alignment satisfies:

$$\left| \langle C, \Gamma^\star \rangle - W_2^2(\hat{r}_{\text{persona}}^\dagger, r_{\text{human}}) \right| = O(\varepsilon). \quad (23)$$

This bound indicates that the OT refinement step achieves fine-grained distribution matching, with an approximation bias that is explicitly controlled by the entropy regularization parameter ε . All proofs are provided in Appendix G.

G.1 Proof of Theorem 1

Theorem 1 (Approximation Error after Importance Sampling). Let $\mathcal{P}^\dagger = \{p_i\}_{i=1}^{N^\dagger}$ be the set of candidate personas selected via importance sampling from an original set of N personas, and let r_{human} denote the empirical human response distribution from M samples. Let $\hat{r}_{\text{persona}}^\dagger$ be the empirical distribution of the responses induced by $\mathcal{P}^\dagger$, and d be the dimension of response vectors. Then with probability at least $1 - \delta$, we have:

$$W_2(\hat{r}_{\text{persona}}^\dagger, r_{\text{human}}) = O\left(h^2 + \sqrt{\frac{\log(1/\delta)}{Nh^d}} + \sqrt{\frac{\log(1/\delta)}{Mh^d}} + \sqrt{\frac{\log(1/\delta)}{N^\dagger}}\right) \quad (24)$$

where $h > 0$ denotes the kernel bandwidth for KDE.

Proof. Denote the *oracle* weights by $w_i^\star = f_{P_h}(x_i)/f_{Q_h}(x_i)$, where $f_{P_h} = f_P * K_h$ and $f_{Q_h} = f_Q * K_h$ are the Gaussian-smoothed densities. Let $w_i^{\text{IS}} = \hat{f}_P(x_i)/\hat{f}_Q(x_i)$ and put $\pi_i^\star = w_i^\star / \sum_k w_k^\star$. Standard KDE bounds (e.g. Tsybakov [62]) give

$$\max_i |w_i^{\text{IS}} - w_i^\star| = O\left(h^2 + \sqrt{\frac{\log(1/\delta)}{Nh^d}} + \sqrt{\frac{\log(1/\delta)}{Mh^d}}\right) \quad (A)$$

with probability $1 - \delta/3$. Because f_{Q_h} is bounded below, dividing by $\sum_k w_k^\star = \Theta(1)$ converts (A) into

$$\|\pi - \pi^\star\|_{\text{TV}} = O\left(h^2 + \sqrt{\frac{\log(1/\delta)}{Nh^d}} + \sqrt{\frac{\log(1/\delta)}{Mh^d}}\right). \quad (B)$$

Since S is compact, TV-Wasserstein duality (15, Lemma 2) implies

$$W_2(\pi, \pi^\star) \leq \text{diam}(S) \sqrt{2 \|\pi - \pi^\star\|_{\text{TV}}}.$$

Combining with (B) yields

$$W_2(\pi, \pi^\star) = O\left(h^2 + \sqrt{\frac{\log(1/\delta)}{Nh^d}} + \sqrt{\frac{\log(1/\delta)}{Mh^d}}\right). \quad (C)$$

By importance-sampling theory $\pi^\star$ is *unbiased* for r_{human} , so $W_2(\pi^\star, r_{\text{human}}) = 0$. Therefore (C) is exactly the *bias* induced by replacing oracle weights with estimated ones.

Conditioned on $\{\pi_i\}$, the multiset $\mathcal{P}^\dagger$ is obtained via $N^\dagger$ i.i.d. draws from π . Applying the concentration bound for empirical Wasserstein distance [15, Theorem 2] gives, with probability $1 - \delta/3$,

$$W_2(\hat{r}_{\text{persona}}^\dagger, \pi) = O\left(\sqrt{\frac{\log(1/\delta)}{N^\dagger}}\right). \quad (D)$$

By a union bound, events (B)-(D) hold with probability at least $1 - \delta$. Then by the triangle inequality and combining (C) and (D), we conclude:

$$W_2(\hat{r}_{\text{persona}}^\dagger, r_{\text{human}}) = O\left(h^2 + \sqrt{\frac{\log(1/\delta)}{Nh^d}} + \sqrt{\frac{\log(1/\delta)}{Mh^d}} + \sqrt{\frac{\log(1/\delta)}{N^\dagger}}\right).$$

Table 10: Seed-persona construction statistics across pipeline stages. “Raw”: original crawled texts. “Preproc”: Preprocessing. “Raw QC”: Raw Text Quality Control. “Persona Gen”: Persona Generation. “Persona QC”: Persona Quality Control.

Source	Raw	Preproc (S1)	Raw QC (S2)	Persona Gen (S3)	Persona QC (S4)
Blog	681,000	500,000	368,000	41,607	30,738

G.2 Proof of Theorem 2

Theorem 2 (Entropic OT Approximation Error). Let $X^\dagger = \{x_i\}_{i=1}^{N^\dagger}$ be the response vectors of personas in $\mathcal{P}^\dagger$, and $Y = \{y_j\}_{j=1}^M$ be the human responses. Let $C \in \mathbb{R}^{N^\dagger \times M}$ be the cost matrix with entries $C_{ij} = \|x_i - y_j\|_2^2$, and let $\Gamma^\star$ be the optimal transport plan obtained via entropic OT with regularization parameter $\varepsilon > 0$. Then the deviation from exact Wasserstein alignment satisfies:

$$\left| \langle C, \Gamma^\star \rangle - W_2^2(r_{\text{persona}}^\dagger, r_{\text{human}}) \right| = O(\varepsilon). \quad (25)$$

Proof. Show that the entropic-OT cost differs from the exact 2-Wasserstein cost by at most $O(\varepsilon)$.

Let

$$\mu = \frac{1}{N^\dagger} \sum_{i=1}^{N^\dagger} \delta_{x_i}, \quad \nu = \frac{1}{M} \sum_{j=1}^M \delta_{y_j},$$

and define the feasible set of uniform couplings $\mathcal{U} = \Pi(\frac{1}{N^\dagger} \mathbf{1}_{N^\dagger}, \frac{1}{M} \mathbf{1}_M)$. Denote the entropic objective

$$\mathcal{S}_\varepsilon(\Gamma) := \langle C, \Gamma \rangle + \varepsilon \sum_{i,j} \Gamma_{ij} (\log \Gamma_{ij} - 1),$$

whose minimiser is $\Gamma^\star$.

Take any optimal un-regularised transport plan $\Gamma^{\text{OT}} \in \mathcal{U}$; then $\langle C, \Gamma^{\text{OT}} \rangle = W_2^2(\mu, \nu)$. Since $\Gamma^\star$ minimises $\mathcal{S}_\varepsilon$,

$$\langle C, \Gamma^\star \rangle + \varepsilon H(\Gamma^\star) \leq W_2^2(\mu, \nu) + \varepsilon H(\Gamma^{\text{OT}}).$$

Dismiss the (negative) entropy on the LHS to obtain

$$\langle C, \Gamma^\star \rangle \leq W_2^2(\mu, \nu) + \varepsilon [H(\Gamma^{\text{OT}}) - H(\Gamma^\star)]. \quad (A)$$

All feasible $\Gamma \in \mathcal{U}$ have entries in $[\frac{1}{N^\dagger M}, 1]$; hence

$$-\log(N^\dagger M) \leq H(\Gamma) \leq 0,$$

so $|H(\Gamma^{\text{OT}}) - H(\Gamma^\star)| \leq \log(N^\dagger M)$. Plugging this into (A) yields

$$\langle C, \Gamma^\star \rangle \leq W_2^2(\mu, \nu) + \varepsilon \log(N^\dagger M). \quad (B)$$

Because Γ^{OT} is feasible for the entropic problem,

$$\mathcal{S}_\varepsilon(\Gamma^\star) \leq \mathcal{S}_\varepsilon(\Gamma^{\text{OT}}) = W_2^2(\mu, \nu) + \varepsilon H(\Gamma^{\text{OT}}),$$

which rearranges (using the same entropy gap) to

$$\langle C, \Gamma^\star \rangle \geq W_2^2(\mu, \nu) - \varepsilon \log(N^\dagger M). \quad (C)$$

By combining (B) and (C), we have:

$$\left| \langle C, \Gamma^\star \rangle - W_2^2(\mu, \nu) \right| \leq \varepsilon \log(N^\dagger M) = O(\varepsilon).$$

Hence

$$\boxed{\left| \langle C, \Gamma^\star \rangle - W_2^2(r_{\text{persona}}^\dagger, r_{\text{human}}) \right| = O(\varepsilon)}.$$

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